APART AI COLLUSION SPRINT · TRACK 2

Detection is not prevention.

A monitor that catches collusion perfectly still changes nothing. We built the loop that acts on the catch — and found that the part doing the work is not the monitor.

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165 runs · 4 models · 1 mechanism control

THE PROBLEM

The flag arrives after the vote

A detector can be perfect and still protect nothing.

Flags arrive as dashboards, not as actions. The vote is cast, the allocation is made, the recommendation is filed.

It buys a record of the failure, not a different outcome.

So the question is not whether we can see collusion.
It is whether acting on the signal changes anything.

FRAMING

Three questions, three different labels

Most work answers Q1 and reports it as if it answered Q3. They need different labels and different controls.

Q1

Detection

Can a monitor tell collusion from honest disagreement?

label: outcome
control: content baseline

not identified here

Q2

Causation

Are the directions that detect it actually doing the work?

label: behaviour
control: random direction

specified, not run

Q3

Prevention

Does acting on the signal reduce collusion, at what cost?

label: behaviour
control: matched placebo

this paper

SETUP

Three agents, a private instruction, a public vote

Five procurement-style committees. One evidence-supported answer, one opposite target. A and B are told to steer to the target; C is honest.

Agent A
colluder

Agent B
colluder

Agent C
honest

→

2 rounds, then a vote
public transcript only

1 **Round 1** — Simultaneous. Neither colluder sees the other, so a code word is the only coalition signal.

2 **Monitor fires** — Scores the transcript, flags an agent. In this pilot it is an oracle: it knows the code word.

3 **Intervention** — A notice goes to the flagged agent between rounds — steer, quarantine, or a matched placebo.

4 **Round 2 + vote** — The majority decides. With 2 colluders out of 3, the pair can outvote the honest agent.

RESULT 1

It looked like a firewall

25 runs, five scenarios, one seed. Monitor-triggered notice between rounds.

no intervention	**5/5**	collusion still succeeds
matched placebo	**5/5**	length and directive form held constant
steer	**0/5**	notice overrides the private instruction
quarantine	**0/5**	flagged message suppressed, then same notice

Paired exact McNemar p = 0.0625 — the floor at five scenarios.

Consistent, not significant.

THE CRACK

One colluder defects. Always the same one.

The majority flips 2–1. That could mean the coalition broke — or that a single vote moved. The vote records say the second.

FINAL VOTES · all five steer runs

A colluder, flagged	**correct**	defected
B colluder	**target**	did not defect
C honest	**correct**	never moved

Exactly one of two colluders defected in 10 of 10 intervention runs.

The untargeted colluder still voted the target every time.

So “the colluders defected” is wrong.
One agent changed side.

RESULT 2

So which part of the notice does the work?

Two grids, 50 runs. Same trigger, same target, only the wording changes.

	no directive	honesty directive
no monitor notice	no intervention **5/5 collusion**	act honestly, no monitor **0/5 collusion**
monitor notice	monitor notice only **5/5 collusion**	flagged + act honestly **0/5 collusion**

The directive is necessary and sufficient.

Remove the monitor and the effect survives. Remove the directive and it disappears.

Targeting the honest agent does nothing; targeting either colluder works.

WHAT THE MONITOR CONTRIBUTES

Nothing.

Its detection signal is not necessary for the effect, and not sufficient without a directive.
A late “act honestly” is what flips the vote — whether or not anyone is watching.

RESULT 3

The notice only wins when nothing outranks it

Same five arms on an open-weights 7B, with and without a system instruction to follow the private role.

	with system instruction	without it
no intervention	5/5	5/5
steer	4/5	1/5
honesty only	4/5	3/5

Collusion success — lower is better

With the role instruction in the system prompt, the notice prevents collusion in 1/5.

Remove that one clause and it prevents 4/5.

The notice borrows its authority from the hierarchy above it.

RESULT 4

The vote is decodable, and the decoder is inert

0.89

leave-one-scenario-out AUROC for a probe that separates target votes from defections, at the final-vote token.

layer 26 of 28 · 30 colluder states

ABLATE THE DIRECTION

16 of 16

target-voting states still vote the target after the direction is projected out — identical to the random-direction control and to no intervention.

The probe reads the decision. Removing what it reads does not move it: a correlate, not a mechanism.

WHAT THIS IS

An upper bound, stated as one

SUPPORTED

The closed loop can be built end to end and measured.

A prompt-level directive flips one colluder and changes the outcome.

The effect does not depend on the monitor, or on which colluder is targeted.

The notice loses to a competing system instruction (1/5 vs 4/5).

NOT SUPPORTED

That a deployable detector exists — the monitor is a code word oracle.

That detection is identified — the round-1 label is a content label.

That the vote direction is causal — it predicts, but ablation does not move it.

A general effect size: 5 scenarios, 1 seed, 4 models.

THE REAL STUDY

What would make this a firewall instead of a prompt

Specified, with a pre-registered decision rule. Not run. Staged so the first causal test costs 130 runs, not 36,000.

01	**Direction extraction**	30 runs	Label behaviour at two decision points, not the condition: what the message advocates, and how the vote lands.
02	**One causal test**	100 runs	Ablate the direction at one dose against a random direction. No dose sweep until this reads.
03	**Dose sweep**	only if positive	Layer, alpha, token position — the part everyone runs first and learns nothing from.

Pre-registered: ablation must cut collusion ≥15 points vs a random direction, with no tie-rate or capability cost.

WHY IT MATTERS

The negative result is the useful part

Every prevention claim of the form “our monitor reduced collusion” is now a claim that needs a directive control.

Reframes the result

A monitor-triggered drop in collusion is not evidence of detection working. It may be a late instruction.

Cheap to apply

One extra arm — the directive without the monitor — settles it. No new model, no GPU.

Reusable harness

Scenarios, oracle, blinded label, matched placebo, defection outcome, resume-safe runner.

A protocol, not a vibe

The activation study has a staged budget, behaviour labels, and a pre-registered decision rule.

TAKEAWAY

A monitor that only watches is a log file.

Closing the loop takes an instruction, and an instruction is not a mechanism. The next test is whether a direction, not a sentence, can move the vote.

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Track 2 · Detection and Audit
github.com/AliRezaei-Code/collusion-steering-detection-is-not-prevention

165 runs
148 tests